

SCOUTIQ

Next-Generation Data-Driven
Football Recruitment Intelligence

FastAPI + PostgreSQL

React + Tailwind

scikit-learn

Divyansh Shrivastava

MSc Sports Analytics | Universidad Europea de Madrid | 2026

Master's Thesis
Final Presentation

992

Players Scored

20

App Pages

40+

Metrics

The Recruitment Problem



Siloed Data Sources

SofaScore, Understat & FotMob have disparate schemas — scouts manually cross-reference spreadsheets, losing hours per player.



No Contextual Scoring

Raw stats treat a CB's tackles the same as a FWD's. Position-blind comparisons lead to poor shortlists and wasted budget.



No Tactical Fit Layer

Clubs cannot assess whether a player's statistical profile matches a specific manager's system demands — until they've already signed.

ScoutIQ Solution: Unified intelligence platform — read-only PostgreSQL data layer, position-weighted composite scores, tactical fit engine, cosine similarity search.

System Architecture



Key Stack Details

DB Pool: size=5, max_overflow=10, pre_ping=ON | pgvector 0.8.0 — IVFFlat index (49ms per query) | 16 routes — zero raw DB errors exposed

Critical constraint: ScoutIQ is strictly READ-ONLY against PostgreSQL. All writes go to SQLite (scouting lists, notes, saved searches).

Data Integration & Coverage

SofaScore

ALL LEAGUES

Primary Source

Match ratings, duels,
tackles, saves, key passes,
shots, accurate passes %

Understat

PL + LA LIGA

Expected Metrics

xG, npxG, xA,
xG_chain, xG_buildup
(PL + La Liga)

FotMob

SELECTIVE

Supplementary

Roster positions, injury
status, market values,
rating cross-validation

Gold View — v_players_current_season

JOIN: players → player_season_stats → teams → leagues → seasons → player_scores

FILTER: is_current = TRUE & minutes ≥ 450

COMPUTE: age = EXTRACT(YEAR FROM AGE(date_of_birth)) — never static

992

Players Scored

40+

Raw Stat Columns

6

League-Seasons

Analytical Engine

Position-Weighted Composite Score

- 1 Raw counting stats fetched from player_season_stats
- 2 Per-90 normalisation: $\text{metric_p90} = (\text{raw} / \text{minutes}) \times 90$
- 3 Position-specific weight vector applied
(GK/CB/FB/CDM/CM/CAM/WNG/SS/CF — 9 roles)
- 4 Percentile rank within (position, season, league) cohort
- 5 0–100 composite score → Elite / Top Tier / Above Average / Average / Developing

Elite 90+

Top Tier 75+

Above Avg 60+

Average 40+

Developing 0+

Cosine Similarity Engine

 $\text{sim}(A,B) =$

$$A \cdot B / (\|A\| \times \|B\|)$$

- ▶ 16-dimensional stat vector (per-90 metrics)
- ▶ L2-normalised, role-weighted before comparison
- ▶ IVFFlat pgvector index — 49ms avg query time
- ▶ 60× speedup over Python cosine (pre-pgvector)
- ▶ Cross-league search via /similar-global endpoint

90%+ Like-for-like replacement 75–89% Stylistic match <70% Different profile

Platform — 20 Application Pages

Dashboard

/

Player Search

/players

Player Profile

/players/:id

Similar Players

/similar

Rankings

/rankings

Scatter Plot

/scatter

Emerging Talent

/talent

Team Style

/team-style

Scouting Lists

/lists

League Coverage

/coverage

Metric Weighting

/weighting

Team Profile

/teams/:id

Chelsea View

/chelsea

Compare

/compare

League Profile

/leagues/:id

Recruitment Board

/recruitment

Tactical Fit

/tactical

Squad Gap

/squad-gap

Profile Shortlist

/profiles/shortlist

Squad Planner

/planner

SECTION II

Chelsea FC Case Study

The Xabi Alonso Era — 2026/2027

 Data bounded to Matchweek 38, 2025/26 PL season. No projected data used.

Jackson (CF)
Palmer (#10) P. Neto (#10)
Kerkez Xhaka Caicedo Gusto
Colwill Palestra Fofana
Sánchez

Alonso System DNA

- ▶ 3-4-2-1 formation
- ▶ Ball-playing CB trio
- ▶ Press-resistant double pivot
- ▶ Wing-backs as width providers
- ▶ Dual #10s in half-spaces
- ▶ High defensive line
- ▶ Possession-dominant build-up

Squad Audit — Retain vs. Sell

✓ RETAINED — Core Squad (9 Players)

- 91 Cole Palmer**
R #10 Franchise Player — Elite xG/xA/KP
- 82 Moisés Caicedo**
RCM 91.2% pass acc, 6.3 rec/90
- 78 Levi Colwill**
LCB 89.4% pass acc — progressive CB
- 77 Reece James**
RWB Club captain — 2.4 dribbles/90, 1.3 KP/90
- 78 Pedro Neto**
L #10 / WNG 3.2 dribbles/90 — direct winger profile
- 74 Wesley Fofana**
RCB 67.8% aerial win, 87.1% pass acc
- 71 Nicolas Jackson**
CF Recalled from Bayern loan — pressing false 9
- 75 Enzo Fernández**
Rotation Pivot rotation — progressive passes
- 72 Robert Sánchez**
GK 73.1% save %, 78.2% distribution acc

✗ SOLD — Departed 25/26 Season (2 Players)

Marc Cucurella — Case Study

Metric	Cucurella	Target	Gap
Successful Dribbles/90	0.7	> 1.5	-53%
Key Passes/90	0.6	> 1.0	-40%
Final Third Passes/90	1.2	> 2.0	-40%
Tackles/90	2.8	> 2.0	+40%

Chalobah

Pass acc 82.8% — below 87% CB threshold for Alonso system

Target Acquisitions — Data-Driven

Alessandro Palestra

Atalanta BC → Chelsea

Libero — CB Anchor

84

Score

23

Age

€55M

Fee

Pass Acc %: **91.3%** 96th PL DEF

Long Balls/90: **3.1** 92nd

Aerial Win %: **68.2%** 82nd

Prog Passes/90: **4.8** 94th

82.3% similar to Colwill — confirmed pair compatibility

Granit Xhaka

Leverkusen → Chelsea

Alonso's Midfield Conductor

81

Score

33

Age

€18M

Fee

Pass Acc %: **92.4%** 97th Bund. MID

Dispossessed/90: **0.3** 12th
(lower=better)

Prog Passes/90: **5.1** 91st

Recoveries/90: **5.8** 82nd

Pre-existing Alonso relationship — minimal integration time

Milos Kerkez

Bournemouth → Chelsea

Attacking Width Provider

77

Score

22

Age

€42M

Fee

Dribbles/90: **2.1** +200% vs Cucurella

Key Passes/90: **1.4** +133% vs Cucurella

Final Third/90: **2.4** +100% vs Cucurella

Tackles/90: **2.6** Defensive parity ✓

Scatter quadrant: top-right (above median dribbles AND key passes)

2026/27 Starting XI — 3-4-2-1



Why This XI Dominates

89.3%

Build-Up Security

avg pass acc across 3 CBs — highest in PL

12.1

Midfield Lockout

combined recoveries/90 (Xhaka + Caicedo)

0.75 xA

Creative Overload

per 90 from 4 attacking-band players

25.4

Squad Age

avg age — sustainable multi-season build

~€72M

Net Spend

€115M in / ~€43M out — focused investment

Transfer Ledger & Financial Summary

IN

Player	From	Role	Fee	Key Data Justification
Alessandro Palestra	Atalanta BC	CB — Libero / Build-up Anchor	€55M	91.3% pass acc (96th pct), 3.1 long balls/90
Milos Kerkez	AFC Bournemouth	LWB — Width Provider	€42M	2.1 dribbles/90 — 200% above Cucurella
Granit Xhaka	Bayer Leverkusen	LCM — Press-Resistant Controller	€18M	92.4% pass acc, 0.3 dispossessed/90

OUT

Player	To	Fee	Data Justification for Sale
Marc Cucurella	TBD	~€28M	Dribbles/90 0.7 vs 1.5 WB target — archetype mismatch (score drops 65→48 when WB-weighted)
Trevoh Chalobah	TBD	~€15M	Pass acc 82.8% — below 87% CB threshold for Alonso high-line system

TOTAL IN: €115M **TOTAL OUT:** ~€43M recovered + wage savings **NET SPEND:** ~€50M — self-funded tactical rebuild

Key Analytical Insights

01

Composite Scores Require Positional Context

Cucurella scored 65 on the generic composite — 'Above Average'. Weighted for the Alonso WB archetype, the score dropped to 48. Raw scores without positional weighting lead to wrong decisions.

02

Cosine Similarity Validates Pair Compatibility

Palestra returned 82.3% similarity to Colwill — confirming the new CB pair would share statistical DNA. The Similarity Engine is a partnership validator, not just a replacement finder.

03

Per-90 vs Raw Totals Changes the Leaderboard

Players with fewer minutes (injury-affected seasons) rank higher per-90. The Rankings page toggle revealed several undervalued players hidden by low raw totals.

04

Scatter Plot Reveals Market Inefficiencies

Kerkez sat in the top-right quadrant (above median dribbles AND key passes) — one of only 4 PL fullbacks there. Bournemouth's market position means he was priced at ~€42M vs a comparable player elsewhere at €80M+.

05

K-Means Cluster Confirms Tactical Compatibility

Leverkusen's Team Style cluster (Possession-Based) matched Chelsea's cluster under Alonso's demanded style. Transfers between stylistically similar clubs show faster adaptation.

06

xG Chain Metric Identifies Systemic Contributors

Traditional stats miss players who build up play without scoring. xG Chain reveals Jackson's systemic contribution (1.1 PWA third/90) — he drives play even when the goals don't come.

SCOUTIQ

Intelligence-Driven Recruitment

- ▶ React 18 + FastAPI + PostgreSQL 15 + pgvector
- ▶ 992 players | 6 league-seasons | 40+ metrics
- ▶ 20 pages | Docker Compose | Production-grade logging
- ▶ Cosine similarity: 49ms avg (60× pre-pgvector speedup)

Divyansh Shrivastava

MSc Sports Analytics — Universidad Europea de Madrid |
github.com/divvyansh05

Platform Summary

20

Application Pages

992

Players Scored

9

Tactical Roles

49ms

Similarity Query

60×

pgvector Speedup

247

README Lines

Thank You — Questions?